



Cryptography in Canada: Why It Matters

Cryptography sits at the centre of Canada's digital economy – protecting online banking, CRA portals, provincial healthcare systems, and mortgage platforms. This course is about understanding how Canadian organizations use cryptography to balance **privacy, security, and usability**.



Digital Trust

The foundation of every secure Canadian online service



Banking & Services

CRA, online banking, and government portals all depend on encryption



Beyond Passwords

Security architecture that goes far deeper than a login screen



Simple Way to Think About It: Cryptography is Canada's digital lock and key system.

Why Canada Depends on Encryption

Canadian organizations face growing pressure to protect information. Financial fraud, ransomware, and identity theft continue to increase – while privacy expectations from citizens and customers are rising. Frameworks such as **PIPEDA**, **PHIPA**, and **OSFI expectations** shape how organizations protect data.

Growing Threats

- Financial fraud and identity theft
- Ransomware targeting institutions
- Rising cyber attack sophistication

Encryption Supports

- Confidentiality of personal data
- Trust between parties
- Regulatory compliance
- Business continuity

⚠️ 🇨🇦 **Canadian Example:** Sending data without encryption is like mailing tax documents on an open postcard. A mortgage broker transmitting SINs or banking data cannot rely on passwords alone.



The Canadian Security Question

One of the biggest misconceptions is believing there is a single "best" encryption method. Canadian organizations rarely rely on one technique – instead, security teams design **layered solutions** based on careful risk assessment.

1

What Are We Protecting?

Identify the sensitivity and classification of the data at risk

2

Who Needs Access?

Define user roles, trust boundaries, and access controls

3

What Are the Constraints?

Consider performance, infrastructure, and regulatory limits

4

Design the Architecture

Combine methods to build a layered, resilient security solution

  **Simple Way to Think About It:** Choosing encryption is like choosing the right vehicle for Canadian weather and terrain – a hospital, a bank, and a smart city sensor network may all use different cryptographic approaches.

Symmetric Encryption: AES

AES – the **Advanced Encryption Standard** – is the workhorse of encryption. It uses one shared key, fast processing, and efficient encryption and decryption, making it ideal for large volumes of data across Canadian organizations.



Cloud Storage

Protecting files and documents stored in Canadian cloud environments



Laptops & Devices

Full-disk encryption on enterprise and government endpoints



Database Security

Encrypting sensitive records including financial and health data



VPNs

Securing remote connections for distributed Canadian workforces

-  **Canadian Example:** A mortgage platform storing borrower files may use AES to protect uploaded financial documents. AES is like a secure filing cabinet where both people share the same key.



How AES Works Behind the Scenes

AES transforms readable information into unreadable ciphertext through **mathematical rounds of scrambling and substitution**. The stronger the key, the harder it becomes to guess or break – making AES 256 a common choice in Canadian enterprise environments.

Key Sizes Available

128-bit

Standard strength

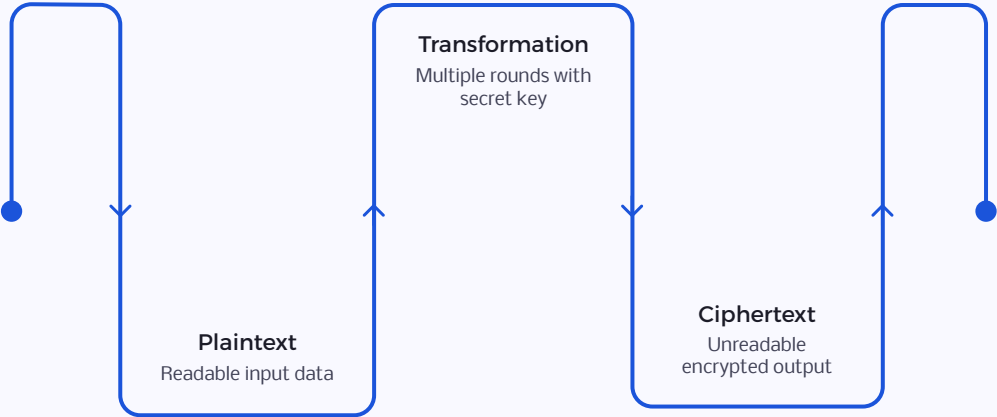
192-bit

Enhanced strength

256-bit

Enterprise grade

The Transformation Process



Each round applies substitution, permutation, and mixing operations to produce unreadable output.



Simple Way to Think About It: Think of scrambling a hockey lineup using a secret rulebook only the coach knows. Encrypted government laptops often rely on AES-based disk encryption.

AES: Strengths and Canadian Challenges

✓ Strengths of AES

Fast

Handles large data volumes with minimal processing overhead

Reliable

Battle-tested standard trusted by governments worldwide

Scalable

Works from individual files to enterprise-wide systems

⚠ The Key Sharing Challenge

AES performs extremely well, but introduces one practical problem: **how do two parties safely share the secret key?**

If someone steals the key:

- Encryption loses all value
- Both sender and receiver become exposed
- Data breaches become inevitable

This is why organizations combine AES with other cryptographic approaches.

⚠ 🗝 **Canadian Example:** A law office emailing clients cannot simply send the AES password in the same email. A strong condo lock means little if the key is taped beside the door.



Asymmetric Encryption and RSA

RSA solved a major encryption problem. Instead of sharing one secret key, it uses a **key pair** – enabling secure communication without first meeting to exchange secrets.

Public Key

Shared openly with anyone who wants to send you encrypted information

Private Key

Protected privately – only the owner can decrypt messages

RSA supports **secure messaging, authentication, and digital signatures** – forming the backbone of trust in Canada's online ecosystem.

  **Simple Way to Think About It:** Anyone can mail a letter to your locked mailbox, but only you can open it. Online banking and secure government portals rely on public key systems.



RSA and Canadian Digital Trust

RSA plays an important role in Canada's online ecosystem. When users visit banks, government sites, or insurance portals, **digital certificates** help verify legitimacy and protect against spoofing, fake websites, and impersonation.



Digital Certificates

Verify the legitimacy of websites and services before users share sensitive data



Authentication

Confirm identity of users and systems in banking and government portals



HTTPS & Trust

Establish encrypted sessions that protect data in transit across Canadian networks



Canadian Example: Secure banking sessions often begin with certificate-based trust using asymmetric cryptography. RSA acts like checking official government ID before entering a secure building.

RSA Limitations

RSA is powerful but **not efficient for encrypting large datasets**. Because of performance overhead, RSA typically handles key exchange and authentication – while AES takes on the heavy lifting of data encryption.

What RSA Handles Well

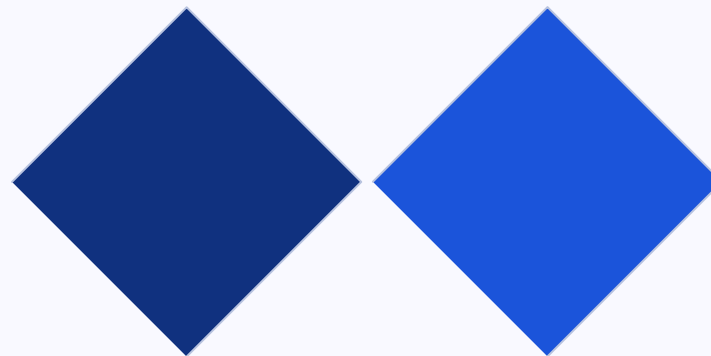
- Session key protection
- Authentication processes
- Digital signatures
- Certificate verification

Why AES Takes Over

- RSA is slower with large data volumes
- Larger key sizes increase processing load
- AES is optimized for bulk encryption
- Combined approach maximizes efficiency

AES Data Encryption

Encrypt large payloads efficiently using session key.



RSA Key Exchange

Authenticate peers and securely deliver session key.

  **Canadian Example:** A Canadian fintech platform may use RSA to establish trust and AES to protect transactions. RSA delivers the safe key rather than moving the entire vault.

Elliptic Curve Cryptography (ECC)

ECC offers the benefits of asymmetric encryption but with **much smaller key sizes** – creating faster performance, lower processing needs, and reduced bandwidth use. This makes ECC increasingly important as Canadian organizations deploy mobile and connected technologies.

256

ECC Key Bits

Equivalent security to a 3072-bit RSA key

10x

Faster

Performance advantage over comparable RSA operations



Lower Overhead

Reduced bandwidth and processing for constrained devices



Simple Way to Think About It: ECC is the compact hybrid car of encryption – smart city projects and connected infrastructure increasingly explore ECC-based security.



ECC and Canada's Connected Future

Canada's connected infrastructure continues to grow – and many of these technologies have **limited power and processing capabilities**. ECC helps maintain strong security while keeping systems efficient and responsive.



Transit Systems

Secure fare collection and passenger data on constrained embedded hardware



Smart Meters

Encrypted energy usage data transmitted from millions of Canadian homes



Environmental Sensors

Lightweight encryption for remote monitoring networks like Durham Region



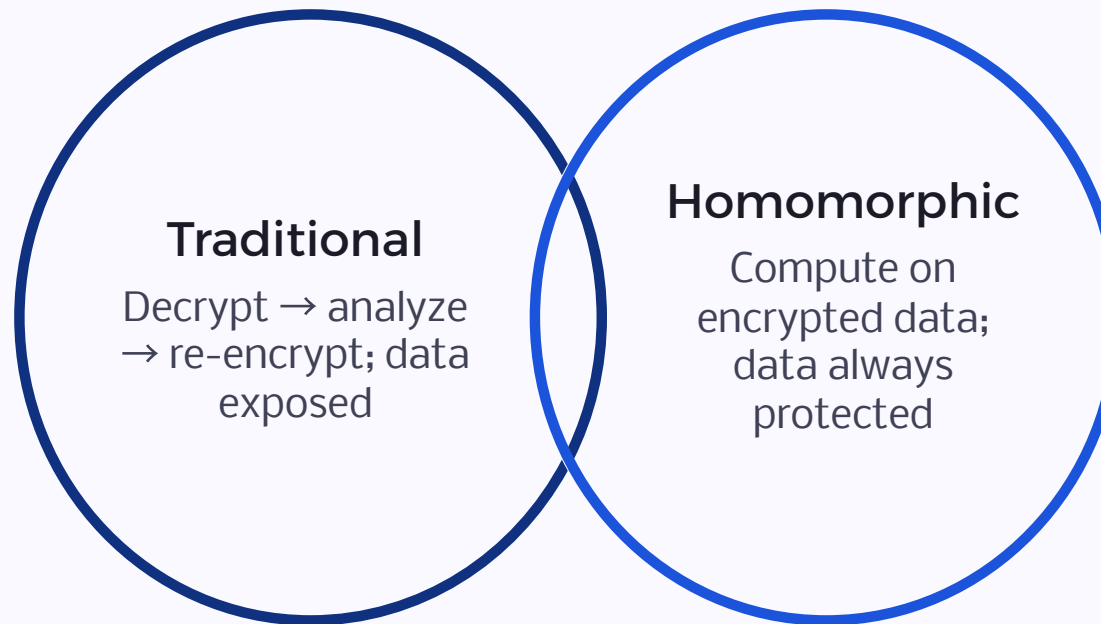
Mobile Health Devices

Protecting patient data on wearables and portable medical equipment

  **Simple Way to Think About It:** ECC is security built for smaller backpacks and lighter equipment – a Durham Region environmental sensor network cannot carry the same encryption overhead as a data centre.

Homomorphic Encryption

Traditional computing requires a risky cycle: **decrypt** → **analyze** → **re-encrypt**. Homomorphic encryption changes this model entirely – allowing organizations to process data and perform calculations while maintaining encryption throughout.



While still computationally intensive, homomorphic encryption represents a **growing and strategically important field** for privacy-preserving analytics in Canada's healthcare and public sectors.

What It Enables

- Process data without decrypting
- Perform calculations on encrypted values
- Maintain privacy throughout analysis

Canadian Applications

- Healthcare analytics and research
- Public sector data aggregation
- Privacy-preserving AI training

📄 🧩 **Simple Way to Think About It:** Imagine solving a puzzle without opening the sealed box holding the pieces.

Paillier and Privacy Analytics

The **Paillier cryptosystem** is a partially homomorphic approach that allows mathematical addition and encrypted aggregation – making it useful when privacy laws or ethics limit data visibility.



✓  **Canadian Example:** A provincial health agency could study population health trends without directly exposing individual patient records. People place numbers into a locked ballot box and only reveal the final total later.

Choosing the Right Encryption in Canada

Canadian cybersecurity professionals do not choose encryption based on popularity. They assess **data sensitivity, privacy obligations, infrastructure limitations, and business needs** – and most real systems combine multiple approaches.



AES → Data Protection

Fast, scalable encryption for files, databases, cloud storage, and VPNs



RSA → Trust & Identity

Authentication, digital certificates, and secure key exchange



ECC → Mobility & Efficiency

Lightweight security for IoT, smart cities, and mobile systems



Homomorphic → Privacy Analytics

Compute on encrypted data for healthcare and public sector insights

  **Canadian Example:** A Canadian bank may use RSA for trust, AES for transactions, and ECC for mobile systems. Security architecture is like building for Canadian weather – one tool is rarely enough.

The Future of Cryptography in Canada

Canada's future digital systems will depend even more on cryptography. The challenge for future professionals is not memorizing formulas – it is understanding how to **engineer trust in complex Canadian systems**.

- 1 — **AI-Driven Attacks**
Machine learning accelerates threat discovery and attack automation
- 2 — **Privacy-Preserving Analytics**
Homomorphic and federated approaches reshape data governance
- 3 — **Quantum Computing Risks**
Post-quantum cryptography standards emerge to protect critical systems
- 4 — **Critical Infrastructure**
Encryption becomes foundational to national resilience and sovereignty

Whether protecting mortgages, hospitals, or smart infrastructure, encryption decisions increasingly shape operational resilience across Canada.

✔  **Simple Way to Think About It:** Cryptography is becoming as essential to Canada as roads, power, and telecommunications.

